INCH-POUND
MIL-D-17650B(SH)
2 December 1988
SUPERSEDING
MIL-D-17650A(SHIPS)
20 April 1966
(See 6.4)

MILITARY SPECIFICATION

DISK, VALVE, RUBBER; INSERT TYPE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers insert type rubber valve disks for sprinkler, void tank flooding in sea water service.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.
MIL-STD-2073-1 - DoD Materiel Procedures for Development and Application of Packaging Requirements.

(Unless otherwise indicated, copies of federal and Military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 395 - Standard Test Methods for Rubber Property - Compression Set. (DoD adopted)
D 412 - Standard Test Methods for Rubber Properties in Tension. (DoD adopted)
D 573 - Standard Test Method for Rubber - Deterioration in an Air Oven. (DoD adopted)
D 2240 - Standard Test Method for Rubber Property - Durometer Hardness. (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The valve disks shall be made of synthetic rubber in sizes as specified (see 6.2).

3.1.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum

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extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Tolerances. The tolerances allowed on the valve disks shall be as specified in table I.

TABLE I. Tolerances.

	Tolerances (inch)	
	Plus	Minus
Outside diameter	0	1/32
Inside diameter	1/32	0
Thickness	1/32	1/32

3.3 Physical requirements. The synthetic rubber in the valve disk shall conform to the physical requirements specified in table II (see 4.4).

TABLE II. Physical requirements of synthetic rubber.

Properties	Requirement	Test method
Initial properties:		
Tensile strength, lb/in^2 minimum	2000	4.4.2
Ultimate elongation, percent, minimum	75	4.4.2
Hardness, durometer points	85 ± 5	4.4.3
Properties after oven aging for 70 ± 1/2 hours at 100 ± 1.1°C (212 ± 2°F):	---	4.4.4
Tensile strength, percent of initial, minimum	80	4.4.4.1
Ultimate elongation, percent of initial, minimum	70	4.4.4.1
Hardness change, durometer points, maximum	5	4.4.4.2
Compression set, percent, maximum	65	4.4.4.3
Properties after immersion for 70 ± 1/2 hours in boiling water:	---	4.4.5
Volume change, percent, maximum (no shrinkage allowed)	15	4.4.5.1
Extraction, percent, maximum	0.5	4.4.5.2
Hardness change, durometer points, maximum	5	4.4.5.3

3.4 Workmanship. The workmanship shall be first class in every respect.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection.

4.2.1 Examination. Each of the samples selected in accordance with 4.3.2 shall be examined for workmanship, dimensions, tolerances, visual defects, and any other requirement of this specification not covered by tests. MIL-STD-407 shall be used to determine and evaluate visual defects.

4.2.1.1 Rejection. All defective or nonconforming disks shall be rejected, and if the number of defective or nonconforming valve disks exceeds the applicable acceptance number of table III, the entire lot represented by the sample shall be rejected.

TABLE III. Sampling for examination.

Number of valve ducts in lot	Number of valve ducts in sample	Number of nonconforming or defective valve disks			
		Major defects[1/]		Total defect[1/]	
		Acceptance number	Rejection number	Acceptance number	Rejection number
0 to 3	All	0	0	1	2
4 to 15	3	0	1	1	2
16 to 50	8	0	1	1	2
51 to 90	13	1	1	1	2
91 to 150	20	1	2	2	3
151 to 280	32	1	2	3	4
281 to 500	50	2	3	4	5
501 to 1200	80	3	4	6	7
1201 and over	125	4	5	8	9

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1/ A major defect is a defect that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit. Total defects are major and minor defects combined.

4.2.2 Production check tests. Production check tests shall be conducted on samples from (or representing) the first lot of material and from every fifth lot thereafter. All the tests specified in 4.4 shall be conducted.

4.2.3 Quality conformance tests. Quality conformance tests shall be conducted on samples from (or representing) all intermediate lots on which production check tests are not performed. The tests specified in 4.4.2, 4.4.3, and 4.4.4.3 shall be conducted.

4.2.4 Action in case of nonconformance. If any of the samples in the production check tests or quality conformance tests is found not to be in conformance with this specification, the lot which it represents shall be rejected. Furthermore, additional samples shall be taken from each subsequent lot and they shall be subjected to the test or tests wherein the failure occurred. Each lot shall be accepted only after satisfactory results are obtained on the test or tests by all the samples taken to represent the lot. This additional testing shall be discontinued after four successive lots have passed the test or tests.

4.3 Sampling.

4.3.1 Lot. For the purposes of sampling, examinations, and tests, a lot shall consist of all valve disks of the same size, produced in one plant under essentially the same conditions, not exceeding 2500 pounds, and offered for delivery at one time.

4.3.2 Sampling for examination. A random sample of valve disks shall be taken from each lot in accordance with the sampling plan given in table III for the examination specified in 4.2.1.

4.3.3 Sampling for tests. Random samples shall be taken from each lot that passes the examination of 4.2.1, in sufficient quantity to conduct either the production check tests or the quality conformance tests specified in 4.2.2 or 4.2.3, as applicable. Since the items are of such size and shape that all test specimens cannot be prepared from them, substitute samples shall be furnished with each lot in the form of three sample pieces 6 inches by 6 inches by 0.080 ± 0.010 inch thick and one sample piece approximately 4 inches by 3 inches by 0.500 ± 0.010 inch thick. The substitute samples shall be certified to be of the same material and equivalent cure as that used in the lot of finished material offered for delivery.

4.4 Test methods.

4.4.1 Testing conditions. Unless otherwise specified herein, the testing conditions shall be in accordance with FED-STD-601.

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4.4.2 Tensile strength and ultimate elongation. The tensile strength and ultimate elongation shall be determined in accordance with ASTM D 412. Die III specimens shall be used for all tensile determinations.

4.4.3 Hardness. Hardness shall be determined in accordance with ASTM D 2240, using a Shore A durometer.

4.4.4 Oven aging. ASTM D 573 shall be used for oven aging the test specimens as specified for the individual tests for 70 ± 1/2 hours at 100 ± 1.1 degrees Celsius (°C) (212° ± 2 degrees Fahrenheit (°F).

4.4.4.1 Tensile strength and elongation after oven aging. The tensile strength and elongation after oven aging shall be determined by the procedure specified in 4.4.2.

4.4.4.2 Hardness after oven aging. The hardness after oven aging shall be determined by the procedure specified in 4.4.3.

4.4.4.3 Compression set after oven aging. Compression set shall be determined in accordance with ASTM D 395, with specimens clamped to 20 percent deflection and aged in accordance with 4.4.4.

4.4.5 Immersion in boiling water. Specimens appropriate for the individual tests, prepared from the valve disks or samples provided in 4.3.3, shall be immersed in boiling 100 ± 1.1°C (212 ± 2°F) distilled water for 70 ± 1/2 hours.

4.4.5.1 Volume change. Volume change after immersion in boiling water shall be determined in accordance with method 6211 of FED-STD-601.

4.4.5.2 Extraction. Extraction after immersion in boiling water shall be determined in accordance with method 6621 of FED-STD-601.

4.4.5.3 Hardness after immersion in boiling water. The hardness of triplicate specimens shall be determined as specified in 4.4.3 before and after immersing the specimens in boiling water for 70 ± 1/2 hours. After the immersion period, the specimens shall be conditioned for 16 to 96 hours at room temperature before testing.

4.5 Inspection of packaging. The inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

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5.1 General.

5.1.1 Navy fire-retardant requirements.

(a) Treated lumber and plywood. When specified (see 6.2.1), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(b) Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packing forms shall meet the flame-spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Preservation. Preservation shall be level A, C or commercial as specified (see 6.2.1).

5.2.1 Level A. Disks shall be unit protected to meet the requirements of MIL-P-116, method III. Quantity per interior (unit and intermediate) packs shall be as specified (see 6.2.1). Interior containers shall be water-resistant folding, set-up, or metal edged paperboard or fiberboard boxes meeting the container requirements of MIL-STD-2073-1, appendix F, table I. Container selection shall be at the option of the contractor.

5.2.2 Level C. Disks shall be unit protected as specified under level A except that the interior containers shall be as follows:

(a) The paperboard containers shall be of the domestic or non-weather resistant type, class, or variety as applicable and,
(b) The fiberboard containers shall be of the class-domestic/fire-retardant material (see 5.1.1(b)). The box closure shall be in accordance with method I using pressure sensitive adhesive tape.

5.2.3 Commercial. Commercial preservation (cleaning, preservation cushioning, unit and intermediate package) shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2.1).

5.3.1 General requirements for levels A, B, and C. Containers selected (see 5.3.2) shall be of minimum weight and cube consistent with the protection required, and of uniform size, and contain identical quantities of identical disks.

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5.3.2 Level A, B, and C containers. Disks preserved as specified (see 5.2), shall be packed in exterior shipping containers for the level of packing specified (see 5.3), in accordance with table VII of MIL-STD-2073-1, appendix C, and herein. Unless otherwise specified (see 6.2.1), container selection shall be at the contractor's option.

5.3.2.1 Caseliners, closure and gross weight.

5.3.2.1.1 Caseliners. Unless otherwise specified (see 6.2.1), level A shipping containers containing disks preserved level C or commercial shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic or fire-retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.3 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.3 Commercial. Disks preserved as specified (see 5.2), shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall have a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid or sill type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage and storage.

5.4 Marking, levels A, B, C, and commercial. In addition to any special marking required (see 6.2.1), interior packs and shipping containers shall be marked including bar coding and the material cure date (quarter and year) for shipment, stowage, and storage in accordance with MIL-STD-2073-1, appendix F.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valve disks covered by this specification are intended for use in sprinkler, void tank flooding in seawater service.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of the specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
(c) Size of valve disk required (see 3.1).

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(d) When fire-retardant treated lumber and plywood is required (see 5.1.1(a)).
(e) Level of preservation and packing required (see 5.2 and 5.3).
(f) Quantity required per each interior pack (see 5.2.1).
(g) Container selection and other options if other than the contractor's (see 5.2.1 and 5.3.2).
(h) When caseliners are not required (see 5.3.2.1.1).
(i) Special marking required (see 5.4).

6.3 Subject term (key word) listing.

Sprinkler
Synthetic
Void tank

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4820-N048)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-17650B(SH)	DISK, VALVE, RUBBER; INSERT TYPE

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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